

cloud computing...

1. Retail Analytics and Recommendation:

Ingest online and offline data via Apache.

Kafka into a delta lakehouse to build unified customer profiles. Use Apache Spark and Redis to serve real-time personalized recommendations directly to user apps.

2. Distributed computing in Research:

Distributed models split massive scientific datasets across network nodes to bypass single-machine CPU / RAM limits. Parallel processing runs simultaneous compute tasks, drastically cutting runtime and enabling seamless horizontal scaling.

3. Real-Time Fraud Detection:

Stream transaction through Apache Flink / Kafka. To run predictive ML models and graph analytics against historical behaviour. Anomalous patterns are detected in milliseconds, allowing the system to block fraudulent payments before settlement.

1) Multimedia Data Management:

Unstructured media region causes high storage overhead and indexing bottlenecks, best solved using distributed object stores like AWS S3. Processing is optimized via distributed GPU pipelines for deep learning.

2) Disaster Recovery & Continuity:-

Implement multi-region active-passive or active-active data application across distinct cloud availability zones. Automated health checks and DNS failover redirect traffic instantly during outages, ensuring near-zero RTO and minimal data loss (RPO).

3) Data Quality & Governance:-

Dirty, duplicate, or missing data corrupts machine learning models and produces flawed analytical decisions. Prevent this by enforcing automated data validation pipelines, strict schema controls, and metadata tracking catalogs like Apache

Atlas.

7) Streaming Recommendation Systems:

capture real-time user behaviour and process

it using spark, ml lib collaborative filtering and.

vector embeddings. the system dynamically recommends items to user feeds on the fly, maximizing user time and audience retention

8) cloud computing in Big data:-

The cloud provides elastic scaling and

decoupled compute / storage, but carries risks of

recurring operational costs and vendor lock-in. strong

IAM roles and end-to-end encryption mitigate security

risks while maintaining flexible deployment.

9) Real - Time Logistics Analytics:

ingest continuous GPS and IoT telematics

streams into Apache Storm/Kafka streams to

compute. live route optimizations. Dynamic

routing avoids traffic and cutting delivery

delays

105 Ethics & Legal policies:-

unregulated big data risks privacy violations
algorithms discrimination, and heavy legal penalties
under frameworks like GDPR organization must
enforce data and routine AI model bias audits